PHILIP B. LUNDRIGAN

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Department of Electrical and Computer Engineering
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EDUCATION

2018 **University of Utah**, *Ph.D., Computer Science*
Advisors: Sneha Kasera and Neal Patwari
Dissertation Title: Reliable Real-Time Data Upload for Wireless Networks

2012 **Brigham Young University**, *B.S., Computer Engineering*
Dean's List, College of Engineering, 2011
Full Academic Brigham Young Scholarship, 2009-2012

PROFESSIONAL EXPERIENCE

Sep 2025 to Present **Brigham Young University**, *Associate Professor*
Department of Electrical and Computer Engineering

Nov 2018 to Aug 2025 **Brigham Young University**, *Assistant Professor*
Department of Electrical and Computer Engineering

PUBLICATIONS

Book Chapters

1. Ramkiran Gouripeddi, **Philip Lundrigan**, Sneha Kasera, Scott Collingwood, Mollie Cummins, Julio C. Facelli, and Katherine Sward, "Exposure Health Informatics Ecosystem", in Total Exposure Health: An Introduction, CRC Press, 2019.

Journal Articles

2. John D. Beard, Maren M. Ball, Christopher Kitras, and **Philip B. Lundrigan**, "Long-term Validation of a Consumer- vs. Industrial-grade Indoor Radon and Air Quality Monitor at a School in Utah", in Environmental Sensing, Sensors, 2026. https://doi.org/10.3390/s26165230

3. Jessica C. Garwood, Xi Yu, Geoffrey A. Hollinger, **Philip B. Lundrigan**, Atsuhiro Muto, Nicholas R. Rypkema, and Yu She, "Recent advances and opportunities for multi-robot systems in oceanography", in Frontiers in Marine Science, 2026. https://doi.org/10.3389/fmars.2026.1823155

4. Callum E. Flowerday, Ryan Thalman, Matthew C. Asplund, Samuel A. Badstubner, Adam K. Cook, **Philip Lundrigan**, Jaron C. Hansen, "Open Path Spectroscopic Detection of Hydroxyl Radical: A Comparison between Broadband Cavity-Enhanced Absorption Spectroscopy (BBCEAS) and BBCEAS Coupled with a

Fabry–Pérot Interferometer (BBCEAS-FP)", in Analytical Chemistry, 2025. https://doi.org/10.1021/acs.analchem.5c01515

5. Daniel Harman, Karl Knapp, Tyler Sweat, **Philip Lundrigan**, Michael Rice, and Willie Harrison, "Physical Layer Security: Channel Sounding Results for the Multi-Antenna Wiretap Channel", in Entropy, 2023. https://doi.org/10.3390/e25101397

6. Callum E. Flowerday, **Philip Lundrigan**, Christopher Kitras, Tu Nguyen, Jaron C. Hansen, "Utilizing Low-Cost Sensors to Monitor Indoor Air Quality in Mongolian Gers", in Low-Cost Sensor Applications for Mobile and Urban Environment Monitoring, Sensors, 2023. https://doi.org/10.3390/s23187721

7. Shruti Hegde, Kyeong T. Min, James Moore, **Philip Lundrigan**, Neal Patwari, Scott Collingwood, and Kerry E. Kelly, "Household Indoor Particulate Matter Measurement Using a Network of Low-Cost Sensors", in Aerosol and Air Quality Research, 2020. https://doi.org/10.4209/aaqr.2019.01.0046

8. James Moore, Pascal Goffin, **Philip Lundrigan**, Neal Patwari, Katherine Sward, J. Weise, Miriah Meyer, "Managing In-home Environments Through Sensing, Annotating, and Visualizing Air Quality Data", in Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), 2018. https://doi.org/10.1145/3264938

9. Tim Strayer, Samuel Nelson, Amando Caro, Joud Khoury, Bryan Tedesco, Olivia DeRosa, Carsten Clark, Kolia Sadeghi, Michael Matthews, Jake Kurzer, **Philip Lundrigan**, Vikas Kawadia, Dorene Ryder, Keith Gremban, Wayne Phoel, "Content Sharing with Mobility in an Infrastructure-less Environment", in Computer Networks, 2018. https://doi.org/10.1016/j.comnet.2018.07.021

10. Brad Mager, **Philip Lundrigan**, and Neal Patwari, "Fingerprint-Based Device-Free Localization: Performance in Changing Environments", in IEEE Journal on Selected Areas in Communications, 2015. https://doi.org/10.1109/jsac.2015.2430515

Conference Publications

11. Joshua Montierth, **Philip Lundrigan**, "WiP: UnderTone: Frequency Lock Encoding on COTS Wi-Fi Hardware", in International Conference on Mobile Computing and Networking (MobiCom), Work-In-Progress (WiP) Program, 2026. https://doi.org/10.1145/3795866.3844741

12. Ethan Sorensen, Ty Young, Brian Smith, **Philip Lundrigan**, "WiP: SAFEC: Slack-Aware Forward Error Correction", in International Conference on Mobile Computing and Networking (MobiCom), Work-In-Progress (WiP) Program, 2026. https://doi.org/10.1145/3795866.3844750

13. Joshua Montierth, Michael Rice, **Philip Lundrigan**, "Frequency Lock Encoding", in IEEE MILCOM, 2026.

14. Ethan Sorensen, Doug Hales, Christopher Kitras, Jamison Lofthouse, Joshua Montierth, **Philip Lundrigan**, "If You Build It, Will They Connect? 6 GHz WiFi in a Stadium", in IEEE International Conference on Computer Communications and Networks (ICCCN), 2026.

15. Derek Benham, Clayton Smith, Tristan Hodgins, Ashton Palacios, **Philip Lundrigan**, Joshua G. Mangelson, "One-Way Acoustic Signal Localization Using Received Signal Strength", in OCEANS 2025 - Great Lakes, 2025. https://doi.org/10.23919/OCEANS59106.2025.11245189

16. Daniel Harman, Ashton Palacios, **Philip Lundrigan**, Willie K. Harrison, "Side Channel Communications With Ghost Modulation Over the Asymmetric Binary Crossover Erasure Channel", in Allerton Conference on Communication, Control, and Computing, 2025. https://hdl.handle.net/2142/130251

17. Christopher Kitras, John D. Beard, James D. Johnston, **Philip Lundrigan**, "Sustainable Radon Mitigation through Optimized HVAC Scheduling", in IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), 2025. https://doi.org/10.1145/3721201.3721398

18. Ashton Palacios, Daniel Harman, Christopher Kitras, Elle Kelsey, Mitchell Burnett, Willie K. Harrison, **Philip Lundrigan**, "Hidden in Plain Sight: Communicating using Interference", in IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), 2025. https://doi.org/10.1109/DySPAN64764.2025.11115916

19. Bryson Schiel, Joshua Montierth, Eli Blattner, Corey E. Dobbs, Tracianne B. Nielsen, **Philip Lundrigan**, "Using Orthogonal Chirps Underwater for In-Band, Full-Duplex Communication with Minimal Self-Interference Cancellation", in ACM International Conference on Underwater Networks & Systems (WUWNet), 2024. https://doi.org/10.1145/3699432.3699464

20. Derek Benham, Ashton Palacios, **Philip Lundrigan**, Joshua Mangelson, "Low-Cost Urban Localization with Magnetometer and LoRa Technology", in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024. https://doi.org/10.1109/IROS58592.2024.10801786

21. Bryson Schiel, Sam Swindler, Alek Farmer, Daniel Sharp, Anup Hassan Murali, Brielle Corry, **Philip Lundrigan**, "A Multi-Layered Framework for Informing V2I Deployment Decisions Using Commercial Hardware-In-The-Loop Testing of RSUs", in IEEE Vehicular Networking Conference (VNC), 2024. https://doi.org/10.1109/VNC61989.2024.10575957

22. Ashton Palacios, Devon Ward, Dinah Bronson, Jon Backman, Deukhyoun Heo, Karl Warnick, **Philip Lundrigan**, "Network Layer Spectral Coordination Integrated With Hadamard Projection for Multilayer Interference Mitigation", in IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), 2024. https://doi.org/10.1109/DySPAN60163.2024.10632837

23. J. Johnson, Ashton Palacios, Cody Arvonen, **Philip Lundrigan**, "Wireless Latency Shift Keying", in International Conference on Mobile Computing and Networking (MobiCom), 2024. https://doi.org/10.1145/3636534.3649373

24. Michael Rice, Hyrum Croft, Joshua Gillis, Zach Hilton, Riley Kirkwood, Preston Walker, **Philip Lundrigan**, Willie Harrison, "A Comparison of Two Software Defined Radios for Aeronautical Telemetry", in Proceedings of the International Telemetering Conference, 2023. http://hdl.handle.net/10150/670459

25. Ashton Palacios, Chase Bledsoe, Elle Kelsey, Laura Landon, Jon Backman, **Philip Lundrigan**, "Stealthy Signals: Using Ghost Modulation to Watermark Interference", in 1st International Workshop on LEO Networking and Communication (LEO-NET), 2023. https://doi.org/10.1145/3614204.3616105

26. Christopher Kitras, Carter Pollen, Kyle Myers, Camille Wirthlin, **Philip Lundrigan**, "Location Verification of Crowd-Sourced Sensors", in IEEE International Conference on Computer Communications and Networks (ICCCN), 2023. https://doi.org/10.1109/icccn58024.2023.10230111

27. Aarushi Sarbhai, Ramkiran Gouripeddi, **Philip Lundrigan**, Pavithra Chidambaram, Aakanksha Saha, Randy Madsen, Julio Facelli, Katherine Sward, and Sneha K. Kasera, "Utilizing a Blockchain for Managing Sensor Metadata in Exposure Health Studies", in Intermountain Engineering, Technology and Computing (IETC), 2022. https://doi.org/10.1109/ietc54973.2022.9796689

28. Lehi Alcantara, Joseph Miera, Batsaikhan Ariun-Erdene, Chia-Chi Teng, **Philip Lundrigan**, "The Hitchhiker's Guide to Successful Remote Sensing Deployments in Mongolia", in Intermountain Engineering, Technology and Computing (IETC), 2020. https://doi.org/10.1109/ietc47856.2020.9249214

29. **Philip Lundrigan**, Neal Patwari, Sneha K. Kasera, "On-Off Noise Power Communication", in International Conference on Mobile Computing and Networking (MobiCom), 2019. https://doi.org/10.1145/3300061.3345436

30. Shobhi Maheshwari, **Philip Lundrigan**, Sneha K. Kasera, "Scheduling Virtual WiFi Interfaces for High Bandwidth Live Video Upstreaming Using Multipath TCP", in International Conference on Distributed Computing and Networking (ICDCN), 2019. https://doi.org/10.1145/3288599.3288620 **Best Paper Award**.

31. **Philip Lundrigan**, Neal Patwari, Sneha K. Kasera, "STRAP: Secure TRansfer of Association Protocol", in IEEE International Conference on Computer Communications and Networks (ICCCN), 2018. https://doi.org/10.1109/icccn.2018.8487333

32. **Philip Lundrigan**, Kyeong T. Min, Neal Patwari, Sneha K. Kasera, Kerry Kelly, James Moore, Miriah Meyer, Scott Collingwood, Flory Nkoy, Bryan Stone, and Katherine Sward, "EpiFi: An In-Home IoT Architecture for Epidemiological Deployments", in IEEE Workshop on Practical Issues in Building Sensor Network Applications (SenseApp), 2018. https://doi.org/10.1109/lcnw.2018.8628482 **Best Paper Runner-up**.

33. Kyeong T. Min, **Philip Lundrigan**, and Neal Patwari, "Smart Home Air Filtering System: A Randomized Controlled Trial for Performance Evaluation", in IEEE/ACM International Conference on Connected Health (CHASE), 2018. https://doi.org/10.1016/j.smhl.2018.07.009

34. **Philip Lundrigan**, Mojgan Khaledi, Makito Kano, Naveen Dasa Subramanyam, and Sneha K. Kasera, "Mobile Live Video Upstreaming", in International Teletraffic Congress (ITC 28), 2016. https://doi.org/10.1109/itc-28.2016.124

35. Randy Buck, Rich Lee, **Philip Lundrigan**, and Daniel Zappala, "WiFu: A composable toolkit for experimental wireless transport protocols", in IEEE International Conference on Mobile Ad-Hoc and Sensor Systems (MASS), 2012. https://doi.org/10.1109/mass.2012.6502529

36. Christopher Lavin, Marc Padilla, Jaren Lamprecht, **Philip Lundrigan**, Brent Nelson, and Brad Hutchings, "HMFlow: Accelerating FPGA Compilation with Hard Macros for Rapid Prototyping", in IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), 2011. https://doi.org/10.1109/fccm.2011.17

37. Christopher Lavin, Marc Padilla, Jaren Lamprecht, **Philip Lundrigan**, Brent Nelson, and Brad Hutchings, "RapidSmith: Do-It-Yourself CAD Tools for Xilinx FPGAs", in International Conference on Field Programmable Logic and Applications (FPL), 2011. https://doi.org/10.1109/fpl.2011.69

38. Christopher Lavin, Marc Padilla, **Philip Lundrigan**, Brent Nelson, and Brad Hutchings, "Rapid prototyping tools for FPGA designs: RapidSmith", in International Conference on Field-Programmable Technology (FPT), 2010. https://doi.org/10.1109/fpt.2010.5681429

Poster/Demo Presentations

39. Jason Rao, Christopher Kitras, and **Philip Lundrigan**, "TCAM: Traffic-Camera-Based Air Quality Monitoring Using Multimodal Large Language Models", in Utah AI Convergence, University of Utah, 2026.

40. Spencer Larson, Amber Allen, Kirsten Steele, Zachary Driskill, Dan Barry, Ethan Sorensen, Nathan Williams, Darrell Sonntag, Matthew R. Jones, **Philip Lundrigan**, and Dale Tree, "Low-Cost Emissions Monitoring for Identifying High Emitters", in Air Quality: Science for Solutions, 2026.

41. Zachary Driskill, Ethan Sorensen, Dan Barry, Sam Swindler, Spencer Larson, Amber Allen, Matthew R. Jones, Darrell Sonntag, Dale Tree, and **Philip Lundrigan**, "Modular, Low-Cost, On-Road Emissions Measurement System", in Air Quality: Science for Solutions, 2026.

42. Spencer Larson, Amber Allen, Kirsten Steele, Zachary Driskill, Dan Barry, Ethan Sorensen, Nathan Williams, Darrell Sonntag, Matthew R. Jones, **Philip Lundrigan**, and Dale Tree, "Low-Cost Emissions Monitoring for Identifying High Emitters", in 36th CRC Real World Emissions Workshop, 2026.

43. Zachary Driskill, Ethan Sorensen, Dan Barry, Sam Swindler, Spencer Larson, Amber Allen, Matthew R. Jones, Darrell Sonntag, Dale Tree, and **Philip Lundrigan**, "Modular, Low-Cost, On-Road Emissions Measurement System", in 36th CRC Real World Emissions Workshop, 2026.

44. Spencer Larson, Amber Allen, Zachary Driskill, Kirsten Steele, J. Gates, Nathan Williams, Dan Barry, Ethan Sorensen, Matthew R. Jones, Darrell Sonntag, **Philip Lundrigan**, and Dale Tree, "Heavy-Duty Diesel Emissions Monitoring Using Low-Cost Sensors", in AGU Annual Meeting, 2025.

45. Jason Rao, Christopher Kitras, and **Philip Lundrigan**, "T.R.A.C. Air: Traffic-camera Real-time Analysis for Clean Air", in For the Benefit of the World Conference, BYU, 2025.

46. Spencer Larson, Amber Allen, Zachary Driskill, Sam Swindler, J. Gates, Matthew R. Jones, Darrell Sonntag, **Philip Lundrigan**, and Dale Tree, "Heavy-Duty Diesel Emissions Monitoring Using Low-Cost Sensors", in 35th CRC Real World Emissions Workshop, 2025.
47. Spencer Larson, Amber Allen, Zachary Driskill, Sam Swindler, J. Gates, Matthew R. Jones, Darrell Sonntag, **Philip Lundrigan**, and Dale Tree, "Heavy-Duty Diesel Emissions Monitoring Using Low-Cost Sensors", in Air Quality: Science for Solutions, 2025.
48. Bryson Schiel, **Philip Lundrigan**, "Role-Based Network Addressing for Fleets of Autonomous Underwater Vehicles", in ACM International Conference on Underwater Networks & Systems (WUWNet), 2024. **Best Poster Award**.
49. Christopher Kitras, Ashton Palacios, **Philip Lundrigan**, "SSS: Building a Seven Segment Sign", in PyCon, 2023.
50. Bryson Schiel, Alek Farmer, Anup Hassan Murali, Brielle Corry, and **Philip Lundrigan**, "Informing V2I Deployment Decisions Using Commercial Hardware-in-the-loop Testing", in IEEE Vehicular Networking Conference (VNC), 2023. https://doi.org/10.1109/VNC57357.2023.10136347
51. Christopher Kitras, Carter Pollan, Kyle Myers, Camille Wirthlin, **Philip Lundrigan**, "Location Monitoring Framework for Citizen Science Sensors", in Air Quality: Science for Solutions, 2023. **Best Poster Award**.
52. Callum E. Flowerday, Ryan Thalman, Matthew C. Asplund, Samuel A. Badstubner, Adam K. Cook, **Philip Lundrigan**, Jaron C. Hansen, "Detection of Ambient Concentrations of Hydroxyl Radical using BBCEAS", in Atmospheric Mechanisms Conference (ACM), 2022.
53. **Philip Lundrigan**, Ramkiran Gouripeddi, Mollie Cummins, Julio Facelli, and Katherine Sward, "Materializing the Air Quality Exposome: The Center of Excellence for Exposure Health Informatics", in Air Quality: Science for Solutions, 2020.
54. **Philip Lundrigan**, Derek Hansen, and Chia-Chi Teng, "Developing an Untethered Network of Low-Power Air Quality Sensors", in Air Quality: Science for Solutions, 2019.
55. Kyeong T. Min, **Philip Lundrigan**, and Neal Patwari, "IASA - Indoor Air Quality Sensing and Automation", in ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), 2017. https://doi.org/10.1145/3055031.3055043

TALKS

Invited Talks

1. **Philip Lundrigan**, "Wireless Protocol Adaptability Through Wireless Subprotocols", at BBN, August 2025.
2. **Philip Lundrigan**, "Hacking 802.11", at Vivint, October 2023.
3. **Philip Lundrigan**, "Wireless Protocol Adaptability Through Wireless Subprotocols", at Riverside Research, March 2023.
4. **Philip Lundrigan**, "Wireless Networking Research for Linux", at BYU Linux Club, February 2022.
5. **Philip Lundrigan**, "College of Engineering PhD Forum", at the University of Utah, October 2021.
6. **Philip Lundrigan**, "Open Source: Learning in Public", at BYU IEEE Branch Meeting, September 2021.
7. **Philip Lundrigan**, "STRAP and Beyond", at BYU Networking Club, December 2019.
8. **Philip Lundrigan**, "Networking Research for Linux", at BYU Linux Club, September 2019.

9. Katherine Sward, **Philip Lundrigan**, Ram Gouripeddi, “An Infrastructure for Generating Exposomes: Initial Lessons from the Utah PRISMS Platform”, at the 27th Annual Meeting of the International Society of Exposure Science (ISES), Research Park Triangle, NC, October 2017.
10. **Philip Lundrigan**, “In-Home Real-Time Sensor Networks”, at the 33rd Annual Utah Conference on Safety & Industrial Hygiene, Salt Lake City, UT, October 2016.
11. **Philip Lundrigan**, Mojgan Khaledi, Makito Kano, Naveen D.S., and Sneha K. Kasera, “Mobile Live Video Upstreaming”, at Raytheon BBN, Cambridge, MA, July 2015.

Teaching Experience

Instructor

- **ECEN 224**: Intro to Computer Systems (Winter 2023, Fall 2023, Fall 2024)
- **ECEN 225**: Intro to Computer Systems (Lab) (Fall 2025)
- **ECEN 330**: Intro to Embedded Systems Programming (Fall 2019)
- **ECEN 426**: Computer Networks (Fall 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025)
- **ECEN 526**: Wireless Networks (Winter 2019, Winter 2020, Winter 2021, Winter 2022, Winter 2023, Winter 2024, Winter 2026)
- **ECEN 626**: Exploring Software Defined Radios (Winter 2025)

Course Development

CpE Course Website Template (BYU) - https://github.com/byu-cpe/student_website_template

- Created easy-to-customize template for Computer Engineering course websites.
- Used in ECEn 224, 320, 330, 423, 426, and 427.

ECEN 224: Computer Systems (BYU) - https://byu-cpe.github.io/ecen224/

- Created the course and all curriculum (slides, homework, tests, labs). This class is required for all department majors and has been taught since 2023.
- Created new hands-on labs for students to build a smart doorbell system. Sourced all materials, including Raspberry Pis, during the electronics shortage caused by COVID-19.
- Piloted class with select students to receive feedback on course material.
- Developed recitations to help students learn material outside of lectures and labs.

ECEN 426: Computer Networks (BYU) - https://ecenetworking.byu.edu/426/

- Created course and all curriculum (slides, homework, tests, labs).
- The computer labs teach students practical skills in building networked applications
- Class covers broad range of topics including data communication reliability, network fairness, security, and privacy.

ECEN 526: Wireless Networking (BYU) - https://ecenetworking.byu.edu/526/

- Created course and all curriculum (slides, homework, tests, labs).

- Students get a combination of exposure to wireless networking research through reading research papers and hands-on experience through labs.

Light The Y (BYU STEM Camp) - https://y-board.github.io/light-the-y/

- Created an interactive circuit board and set of lessons to teach high school students how to program.
- Developed a second version Y-Board circuit board and accompanying software libraries.
- The curriculum has been used in BYU's STEM Camp in 2023, 2024, and 2025.

ECEn 192: Student Project (BYU) - https://byu-cpe.github.io/ecen192/

- Worked with students to help develop Y-Board portion of ECEn 192 programming labs.

Y-Board Mini - https://github.com/y-board/mini-hardware

- Created Y-Board Mini to teach new students how to solder.
- Built a custom solder jig and programmer to make the experience scalable and approachable for students with no prior soldering experience.
- Has been used by middle school and high school students, including participants in BYU's CHIP camp.

AWARDS

2025 **NSF CAREER Award**
Using Protocol Blind Spots to Develop Opportunistic Wireless Subprotocols for Advanced Adaptability

2024 **WUWNet 2024 Best Poster Award**
For "Role-Based Network Addressing for Fleets of Autonomous Underwater Vehicles"

2022 **NSF Ideas Lab: Engineering Technologies to Advance Underwater Sciences (ETAUS) participant**

2021 **Most Influential Faculty Member Award, Electrical and Computer Engineering**

2019 **ICDCN Best Paper Award**
For "Scheduling Virtual WiFi Interfaces for High Bandwidth Live Video Upstreaming Using Multipath TCP"

2018 **SenseApp Best Paper Runner-up**
For "An In-Home IoT Architecture for Epidemiological Deployments"

2016 **Best Poster Award, University of Utah, School of Computing Poster Competition**

Citations

Citations: 807
h-index: 10
i10-index: 10

Employment Experience

2012 to 2018 **Research Assistant**, *Advanced Networks Systems Research Lab*
University of Utah

2015, Summer **Network Scientist Intern**, *Raytheon BBN Technologies*
Cambridge, MA

2014, Summer **Network Scientist Intern**, *Raytheon BBN Technologies*
Cambridge, MA

2013 to 2014 **Wireless Researcher**, *Xandem Technology*
Salt Lake City, UT

2012, Summer **Software Engineering Intern**, *Ancestry.com*
Provo, UT

2011 to 2012 **Research Assistant**, *Internet Research Lab*
Brigham Young University

Professional Activities

Senior Member of IEEE and IEEE ComSoc

Member of ACM and ACM SIGMOBILE

Member of Center of Excellence for Exposure Health Informatics at the University of Utah

Utah AI.Summit Judge - 2025

NSF Panel Reviewer - 2022, 2024

University of Utah Sensor Expert Panel - 2024, 2025

DARPA ISAT Study Group Participant - 2024

Organizing Committee
Air Quality: Science for Solutions (Local Utah Conference) - 2019 to present

Poster and Demo Co-Chair
IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN) - 2025

Publicity Co-Chair
IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM) - 2020

Technical Program Committee (TPC) Member
ACM Conference On Mobile Computing And Networking (MobiCom) - 2026, 2027
IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN) - 2025, 2026
ACM Conference On Mobile Computing And Networking (MobiCom) Artifact Committee - 2024
ACM/IEEE International Conference on Internet of Things Design and Implementation (IoTDI) - 2022, 2023, 2024
IEEE International Conference on Local Computer Networks (LCN) - 2021, 2022, 2023, 2024, 2025
ACM Conference on Embedded Networked Sensor Systems (SenSys) - 2023, 2024, 2025
IEEE Vehicular Technology Conference (VTC) - 2023
ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN) - 2020
IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM) - 2020

Session Chair
ACM/IEEE International Conference on Internet of Things Design and Implementation (IoTDI) - 2023
International Symposium on a World of Wireless, Mobile, and Multimedia Networks (WoWMoM) - 2020

Reviewer
IEEE Transactions on Mobile Computing - 2020

John A. Widtsoe Grant Reviewer
Represented the College of Engineering reviewing grant proposals.

College of Engineering CAEDM Manger Hiring Committee (2022)
Represented department in interviewing a new CAEDM manager for the college.

BYU ECEn Department Scheduler (2019 to Present)
In charge of finding times and classrooms for all courses and labs in department. Work with college scheduler and university schedulers to determine availability classrooms. Work with faculty to account for teaching needs and preferences.

BYU ECEn Undergraduate Committee (2018 to Present)
Meet every other week as a committee to discuss needs of undergraduates in department. Helped with merging IT&C major into our department.

College of Engineering Teaching Committee (2021)
Helped develop tools and best practices for remote teaching during COVID-19. Helped to organize a virtual lunch and learn meeting for college faculty members.

GRANTS

2026 to 2027 Darrell Sonntag, **Philip Lundrigan**, Matthew Jones, Dale Tree
"Exploring Smart and Low-Cost Monitoring Solutions for Cleaner Heavy-Duty Diesel Vehicles"
Utah Clean Air Partnership (UCAIR), $15,000

2026 to 2027 Darrell Sonntag, **Philip Lundrigan**, Matthew Jones, Dale Tree
"Exploring Smart and Low-Cost Monitoring Solutions for Cleaner Heavy-Duty Diesel Vehicles"
Ira A. Fulton College of Engineering, BYU, $12,500

2025 to 2026 **Philip Lundrigan** (PI), DJ Lee, Jaron Hansen, Kevin John
"T.R.A.C. Air: Traffic-camera Real-time Analysis for Clean Air"
BYU Interdisciplinary Research Award, $40,000

2025 to 2026 **Philip Lundrigan** (PI)
"Generating Communication Channels to Operate (GeCCO) Phase 2"
DARPA (through Riverside Research and BBN), $214,617

2025 to 2030 **Philip Lundrigan** (PI)
"CAREER: Using Protocol Blind Spots to Develop Opportunistic Wireless Subprotocols for Advanced Adaptability"
National Science Foundation (NSF), $532,556

2025 to 2026 **Philip Lundrigan** (PI)
"Monitoring Visitors' Hand Washing Compliance in NICUs"
Ira A. Fulton College of Engineering, BYU, $12,500

2024 to 2026 Darrell Sonntag (PI), **Philip Lundrigan** (Co-PI), Matthew Jones (Co-PI), Dale Tree (Co-PI)
"Development of a Heavy-duty Truck Emissions Monitor"
Utah State, $522,541

2023 to 2026 **Philip Lundrigan** (PI)
"Meshed Observations of THE Remote Subsurface with Heterogeneous Intelligent Platforms (MOTHERSHIP)"
National Science Foundation (NSF), $209,608

2023 to 2024 John Beard (PI), **Philip Lundrigan** (Co-PI), James Johnston (Co-PI), Scott Collingwood (Co-PI)
"Sustainable, Low-cost Radon Mitigation Strategies for K-12 Schools and Employees"
Rocky Mountain Center for Occupational and Environmental Health (RMCOEH), $19,840

2023 to 2025 **Philip Lundrigan** (PI)
"Investigating Privacy-Preserving Techniques Using Wireless Sub-Protocols"
DARPA (through Riverside Research), $327,424

2022 to 2023 **Philip Lundrigan** (PI)
"RSE Performance, Interoperability, and ITS Device Integration"
Panasonic Corporation of North America (Smart Mobility Office), $20,000

2022 to **Philip Lundrigan** (PI)

2023 "Using Traffic Cameras to Measure Air Quality"
Ira A. Fulton College of Engineering, BYU, $12,500

2022 to 2024 **Philip Lundrigan** (PI)
"CRII: CNS: Building A Framework for Software-Based Wireless Sub-Protocols"
National Science Foundation (NSF), $171,494

2021 to 2022 Bryan Hopkins (PI), Ruth Kelly (Co-PI), and **Philip Lundrigan** (Co-PI)
"Water Conservation in the Arid West: Spatio-Temporal Analysis and Variable Rate Irrigation in the Urban Environment"
Redd Center, BYU, $5,000

2020 to 2023 Karl Warnick (PI) and **Philip Lundrigan** (Co-PI)
"Spectrum Sharing Via Interference-resilient Passive Receivers and Passive-aware Active Services"
National Science Foundation (NSF), $306,935

2019 to 2020 **Philip Lundrigan**
"Mongolian Ger Air Quality Measurement and Analysis"
Donation from Shipps, $35,775

2019 to 2020 **Philip Lundrigan** (PI)
"Mongolian Ger Air Quality Measurement and Analysis"
Deseret International Charities, $15,250

2019 to 2020 **Philip Lundrigan** (PI)
"Self-Sustainable Air Quality Sensor"
Ira A. Fulton College of Engineering, BYU, $12,500

2018 to 2020 **Philip Lundrigan** (PI)
"PRISMS Informatics Platform - Federated Integration Architecture"
National Institutes of Health (NIH), $21,208